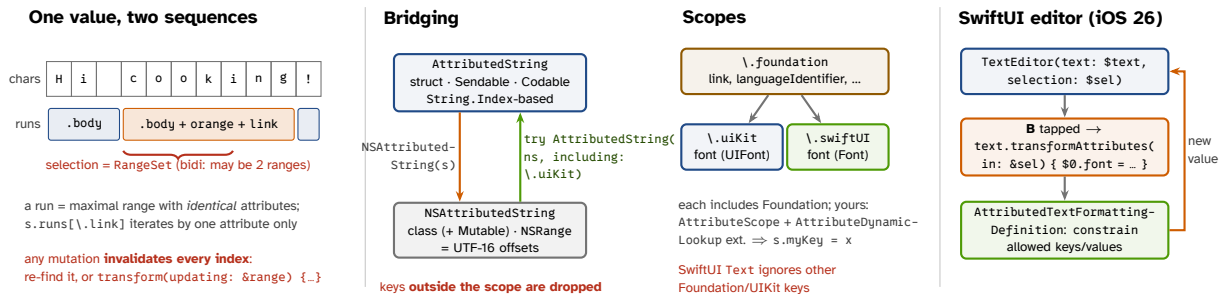


In one line: `AttributedString` (iOS 15) is a Swift **value type**: a sequence of **characters** plus a sequence of **runs**, each run a range with one **AttributeContainer**; attributes are **type-safe keys** grouped in **scopes** (Foundation, UIKit, SwiftUI, yours). `NSAttributedString` is the older reference type keyed by `NSRange`, still what `TextKit` stores. iOS 26 makes it editable in SwiftUI: `TextEditor(text: $attributed, selection: $sel)`.



AttributedString

- Views: characters, unicodeScalars, utf8/utf16, runs. `s[range] → AttributedSubstring; s[rangeSet] → DiscontiguousAttributedString`.
- `s.foregroundColor = .red` sets it on every run; or build an `AttributeContainer → mergeAttributes / replaceAttributes(_with:) / setAttributes`.
- Markdown:** `try AttributedString(markdown:)` — inline styles and links become attributes; blocks become `presentationIntent`, which `Text` does not draw (`.inlineOnlyPreservingWhitespace` keeps newlines). `Text("**hi**")` parses a *literal*; `Text(someString)` does not.
- Custom keys:** `CodableAttributedStringKey` (`MarkdownDecodable...` for `^[text](key: value)`); rules: `inheritedByAddedText`, `invalidationConditions [.textChanged]`, `runBoundaries .paragraph`.
- In SwiftUI, string attributes **beat view modifiers**; SwiftUI-scope keys beat UIKit ones.

AttributedString vs NSAttributedString

	AttributedString	NSAttributedString
kind	struct, COW, Sendable	class; NSMutable_ to edit
keys	typed (\.font), scoped	[NSAttributedString.Key: Any]
index	AttributedString.Index	NSRange (UTF-16 ints)
Codable	yes (codable keys)	NSSecureCoding, RTF(D), HTML
used by	SwiftUI Text/TextEditor	UIKit views, TextKit, Core Text

Example — build, walk, bridge; a Bold button (iOS 26)

```
var s = try AttributedString(
  markdown: "Read the [docs](https://example.com)")
var hot = AttributeContainer(); hot.foregroundColor = .orange
s.mergeAttributes(hot)
for run in s.runs where run.link != nil {
  print(String(s[run.range].characters)) } // "docs"
let ns = NSAttributedString(s) // hand to UIKit
// SwiftUI: @State text: AttributedString, sel: AttributedStringSelection
TextEditor(text: $text, selection: $sel) // B/I/U built in
func toggleBold() { // ctx: fontResolutionContext
  let f = sel.transformAttributes(in: text).font ?? .default
  let on = f.resolve(in: ctx).isBold
  text.transformAttributes(in: &sel) {
    $0.font = ($0.font ?? .default).bold(fon) } }
```

Links & data detectors

`.link` attribute or markdown link; SwiftUI intercepts via `.environment(\.openURL, OpenURLAction { _ in .handled })`. `UITextView`: link colour comes from `linkTextAttributes`, not the run; iOS 17 `textView(_:primaryActionFor:defaultAction:)` gets a `UITextItem` (link, attachment, tag). `dataDetectorTypes` work **only when not editable** — else `NSDataDetector`.

Rich-text editing — SwiftUI (iOS 26)

- `TextEditor(text: $attributed, selection:); system B/I/U` in the context menu + keyboard bar (`.textInputFormattingControlVisibility(_:for:)`).
- `AttributedStringSelection = caret (+ typing attributes)` or ranges: `indices(in:), attributes(in:), typingAttributes(in:)`.
- Mutate *through* the selection so it stays valid: `transformAttributes(in:), replaceSelection(_withCharacters:)`.
- Limit formatting: `AttributedStringFormattingDefinition` (Scope + ValueConstraints) via `.attributedStringFormattingDefinition(_:)` — controls for disallowed values disable themselves.

Rich-text editing — UIKit

- Edit `textStorage` in place (keeps selection + undo), don't reset `attributedString`. `allowsEditingTextAttributes` adds B/I/U; iOS 18 `textFormattingConfiguration` feeds the formatting panel.
- `typingAttributes` style the next typed text; **cleared when the selection changes**. iOS 26 multi-selection: `shouldChangeTextInRanges:replacementText:.`

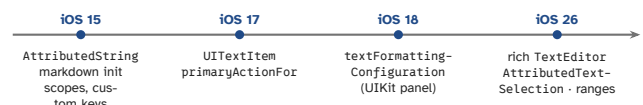
Text input — UITextField & IME

- `UIKeyInput` (`insertText, deleteBackward, hasText`) suffices for a PIN field; a real editor adopts `UITextField` (positions, ranges, geometry, tokenizer, inputDelegate).
- Marked text:** CJK IMEs compose in `markedTextRange` (`setMarkedText → unmarkText`) — not committed text yet.

Traps

- ! Max-length check in `shouldChangeText...` ignoring `markedTextRange` breaks CJK typing — skip while composing.
- ! Keeping an `AttributedString.Index` across an edit — use `transform(updating:)`.
- ! `NSRange` built from `String.count` — UTF-16 ≠ Characters (emoji).
- ! HTML import (`NSAttributedString(data:options:), .html`) spins up WebKit: main thread, slow — never in a cell.

Timeline



Remember

"Characters + runs; scope the keys; edit through the selection; mind the marked text."